

Interpret Division of Fractions

Lesson 2-1

Name:

Date:

Class:

Key Vocabulary

Level 1 support

Picture first, then the word, then a plain-language meaning. Say each word out loud.

12 dots

$$12 \div 3 = 4$$

In $3 \div 1/4 = 12$, the dividend is 3 — it is the total being split

Dividend

The number you are splitting up in a division problem.



$$12 \div 3 = 4$$

In $3 \div 1/4 = 12$, the divisor is $1/4$ — it is the size of each piece

Divisor

The number you split by in a division problem.



$$12 \div 3 = 4$$

In $3 \div 1/4 = 12$, the quotient is 12 — there are 12 quarter-size pieces in 3

Quotient

The answer when you divide.

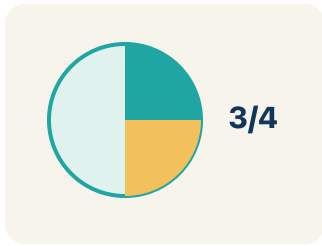
$$3/4 \rightarrow 4/3$$

flip top and bottom

The reciprocal of $1/4$ is $4/1 = 4$. Multiplying by the reciprocal gives the same result as dividing.

Reciprocal

A fraction turned upside down.



$\frac{1}{2}$, $\frac{1}{3}$, $\frac{1}{4}$, $\frac{1}{5}$ — each represents one equal part of a whole

Unit fraction

A fraction with 1 on top, like $\frac{1}{4}$.

Key Ideas & Notes



WHAT WE'RE LEARNING TODAY

I can use models to interpret what it means to divide by a fraction.

1 Notice

Agent Reyes finds a 3-foot strip of tape at the crime scene.

2 Key idea

I can use models to interpret what it means to divide by a fraction.

3 Words I'll use

Dividend

Divisor

Quotient

Reciprocal

Unit fraction



Think About It

- What total length is being divided?
- What is the size of each section?
- Will the answer be greater than or less than 3?

See the notes in action: watch one worked all the way through, then try the next with the same steps.



I do — watch

Follow each step as your teacher solves it.

Problem: What does $3 \div \frac{1}{4}$ mean?

- A. How many $\frac{1}{4}$ -size pieces fit into 3
- B. 3 groups of $\frac{1}{4}$
- C. 3 minus $\frac{1}{4}$
- D. $\frac{1}{4}$ of 3

Step 1 $3 \div \frac{1}{4}$ asks: how many $\frac{1}{4}$ -size pieces fit into 3?

Step 2 The answer is 12, because each whole has 4 fourths, and $3 \times 4 = 12$.



Answer: A. How many $\frac{1}{4}$ -size pieces fit into 3

 We do — together

Solve this one with your class using the same steps.

Problem: Which expression means 'how many $\frac{1}{3}$ -size pieces are in 2'?

A. $2 \div \frac{1}{3}$

B. $2 \times \frac{1}{3}$

C. $\frac{1}{3} \div 2$

D. $2 + \frac{1}{3}$

Step 1

Step 2

Answer:

 You do — your turn

Now try one on your own. Show every step.

Problem: A rope is 4 feet long. How many $\frac{1}{2}$ -foot pieces can be cut from it?

A. 8 pieces

B. 2 pieces

C. 4 pieces

D. $\frac{1}{2}$ piece

Show your work:

Turn & Talk — Discussion Points

Level 1 support

Level 2

Talk with a partner about each prompt. If you need help getting started, use the Level 1 sentence stems. Ready for more? Try the Level 2 question.

LAUNCH

Agent Reyes cuts a 3-foot strip of tape into $\frac{1}{4}$ -foot sections. Before calculating, will the number of sections be more or less than 3? Why?

Level 1 support

Start here: There will be more than 3 sections because each piece is smaller than a whole foot.

 **Need a hint?**

Sentence starters (optional)

The number of sections will be ____ than 3 because ____.

El número de secciones será ____ que 3 porque ____.

Each $\frac{1}{4}$ -foot piece is ____ than one whole foot, so ____.

Cada pieza de $\frac{1}{4}$ de pie es ____ que un pie entero, así que ____.

Word bank: dividend divisor unit fraction

Listen for: Students reason that dividing by a number less than 1 gives a quotient greater than the dividend, so there are more than 3 sections.

Level 2

Predict how the number of sections would change if each piece were $\frac{1}{8}$ foot instead of $\frac{1}{4}$ foot. Justify your prediction.

- With $\frac{1}{8}$ -foot pieces there would be ____ sections because ____.
- A smaller divisor produces ____ because ____.

EXPLORE

What does $3 \div \frac{1}{4}$ actually mean? Show your partner with the bar model.

Level 1 support

Start here: $3 \div \frac{1}{4}$ asks how many $\frac{1}{4}$ -size pieces fit into 3.



Need a hint?

Sentence starters (optional)

Dividing 3 by $\frac{1}{4}$ means finding how many ____ fit into ____.

Dividir 3 entre $\frac{1}{4}$ significa encontrar cuántos ____ caben en ____.

My bar model shows ____ fourths, so the quotient is ____.

Mi modelo de barras muestra ____ cuartos, así que el cociente es ____.

Word bank: **dividend** **divisor** **quotient** **unit fraction**

Listen for: Students name 3 as the dividend and $\frac{1}{4}$ as the divisor, and explain the model shows 12 fourths fit into 3.

Level 2

Why is $3 \div \frac{1}{4}$ NOT the same as $\frac{1}{4}$ of 3? Critique a classmate who confuses the two.

- $3 \div \frac{1}{4}$ is different from $\frac{1}{4}$ of 3 because ____.
- Taking $\frac{1}{4}$ of 3 gives ____, but dividing gives ____.

CONNECT

Where in cooking, sharing, or building might you divide a whole by a fraction?

Level 1 support

Start here: Any time you ask 'how many small pieces fit' you are dividing by a fraction.



Need a hint?

Sentence starters (optional)

In real life I divide by a fraction when ____.

En la vida real divido entre una fracción cuando ____.

For example, ____ uses ____ $\div$ ____.

Por ejemplo, ____ usa ____ $\div$ ____.

Word bank: **dividend** **divisor** **quotient**

Listen for: Students give a 'how many fit' scenario and correctly identify the whole as the dividend and the piece size as the divisor.

Level 2

Generalize: how can you tell from the words of a problem that you should divide by a fraction rather than multiply?

- The clue that tells me to divide is ____.
- If the problem instead asked for ____, I would multiply because ____.

Write About the Math

The Writing Revolution

I can explain a fraction division using the words dividend, divisor, quotient, reciprocal, and unit fraction.

1. Kernel Sentence subject + verb

Model: Dividend is the number you are splitting up in a division problem.

Dividendo es el número que estás repartiendo en una división.

Write a kernel sentence about dividend. Use a subject and a verb.

Escribe una oración base sobre dividendo. Usa un sujeto y un verbo.

2. Sentence Expansion because · but · so

Kernel: Dividend matters in math

Dividendo importa en matemáticas

Expand the kernel three ways. Add a reason, a contrast, and a result.

because
porque

Dividend matters in math because ____.

Dividendo importa en matemáticas porque ____.

but
pero

Dividend matters in math, but ____.

Dividendo importa en matemáticas, pero ____.

so
entonces

Dividend matters in math, so ____.

Dividendo importa en matemáticas, entonces ____.

3. Sentence Types 4 ways to write a math idea

Statement *Afirmación*

Tell one true fact about dividend.
Di un hecho verdadero sobre dividend.

Dividend ____.

Question *Pregunta*

Ask a question about dividend.
Haz una pregunta sobre dividend.

How does ____ ?

¿Cómo ____ ?

Exclamation *Exclamación*

Show excitement about dividend.
Muestra entusiasmo sobre dividend.

Wow, ____ !

¡Guau, ____ !

Command *Mandato*

Tell a partner what to do with dividend.
Dile a un compañero qué hacer con dividend.

First, ____ .

Primero, ____ .

4. Explain Your Reasoning use a sentence starter

The number of sections will be ____ **than 3 because** ____.

El número de secciones será ____ *que 3 porque* ____.

Each 1/4-foot piece is ____ **than one whole foot, so** ____.

Cada pieza de 1/4 de pie es ____ *que un pie entero, así que* ____.

Dividing 3 by 1/4 means finding how many ____ **fit into** ____.

Dividir 3 entre 1/4 significa encontrar cuántos ____ *caben en* ____.

Try It

Solve on your own. Check the answer key when you are done.

1. Which division does this picture show: a length of 3 split into $\frac{1}{4}$ -size pieces?

A. $3 \div (\frac{1}{4}) = 12$

B. $3 \div 4 = \frac{3}{4}$

C. $\frac{1}{4} \div 3$

D. $3 \times 4 = 7$

Show your work:

2. How many $\frac{2}{3}$ -cup scoops are in 4 cups?

A. 6

B. 8

C. 3

D. 2

Show your work:

Stretch Your Thinking

Level 2 enrichment

Challenge task — explain your reasoning in full sentences.

Dividing 6 by $\frac{1}{2}$ gives 12, but dividing 6 by 2 gives 3. How can dividing by a smaller number ($\frac{1}{2}$) give a bigger answer than dividing by a larger number (2)? Explain using a real-world example.

Sentence starter: When I divide 6 by 2, I am _____. When I divide 6 by $\frac{1}{2}$, I am _____. The answer is bigger because _____.

Show your work:

Reflect — Exit Ticket

What does $5 \div \frac{1}{4}$ mean, and what is the quotient?

- A. How many $\frac{1}{4}$ -size pieces fit into 5; quotient is 20
- B. 5 groups of $\frac{1}{4}$; quotient is $\frac{5}{4}$
- C. $\frac{1}{4}$ of 5; quotient is $\frac{5}{4}$
- D. 5 minus $\frac{1}{4}$; quotient is $4\frac{3}{4}$

Your answer:

Answer Key & Teacher Guide

1. **We Try (notes):** A. $2 \div 1/3$ — *Finding how many $1/3$ -size pieces fit into 2 is represented by $2 \div 1/3$. The answer is 6.*
2. **You Try (notes):** A. 8 pieces — $4 \div 1/2 = 4 \times 2 = 8$ pieces. *Each whole foot contains 2 halves, and $4 \times 2 = 8$.*
3. **Try It 1:** A. $3 \div (1/4) = 12$ — *Each whole holds 4 fourths, so 3 wholes hold 12 fourths.*
4. **Try It 2:** A. $6 - 4 \div (2/3) = 4 \times (3/2) = 12/2 = 6$ scoops.
5. **Exit Ticket:** A. How many $1/4$ -size pieces fit into 5; quotient is 20 — $5 \div 1/4$ means 'how many $1/4$ -size pieces fit into 5.' *Since each whole has 4 fourths, $5 \times 4 = 20$.*

Writing (TWR) — what to look for

- **Kernel sentence:** A complete sentence needs a subject and a verb. Example: Dividend is the number you are splitting up in a division problem.
- **Expansion:** *because* gives a reason, *but* shows a contrast or exception, *so* shows a result. Answers vary; each must keep the kernel idea and add the correct kind of detail.
- **Sentence types:** Statement ends with a period, question with "?", exclamation with "!", and a command starts with an action verb (a "bossy" verb).